Explicit bidirectional separation of product and additive integrability

A full answer to the first open problem in arXiv:1410.1845

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Abstract

We construct step mappings with well-ordered steps which separate Kurzweil product integrability and Henstock–Kurzweil integrability in both directions. The construction uses exact five-factor commutator blocks in $M_3(\mathbb{R})$: their exponential product is the identity while their additive sum has a harmonic nilpotent drift. Removing the correcting fifth factor reverses the separation. A principal-log refinement gives the product-integrable but non-Henstock–Kurzweil example already in $M_2(\mathbb{R})$. We also isolate a missing local-log condition in the sequence-level equivalence asserted in the source, and verify the original step-integrability statements directly.

1 Source question and answer

The source is S. Heikkilä and A. Slavík, *On summability, multipliability, product integrability, and parallel translation*, arXiv:1410.1845; J. Math. Anal. Appl. 433 (2016), 887–934. Section 9, PDF page 44, asks:

> Is there a step mapping with well-ordered steps which is Kurzweil product integrable but not Henstock–Kurzweil integrable (or vice versa)?

> ### 9 Open problems
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> We conclude our paper with several open problems:
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> - It is known that already in the finite-dimensional Banach algebra $E = \mathbb{R}^{2\times 2}$, Kurzweil product integrability and Henstock-Kurzweil integrability of a function $A : [a, b] \to E$ do not imply each other (see [11, Example 4.7]). However, the existing examples are not quite elementary. Is there a step mapping with well-ordered steps which is Kurzweil product integrable but not Henstock-Kurzweil integrable (or vice versa)? Equivalently, is there a well-ordered set $\Lambda \subset [a, b]$ and a family $(x_\alpha)_{\alpha\in\Lambda}$ such that $(\exp x_\alpha)_{\alpha\in\Lambda^{<b}}$ is multipliable and its product is invertible, but $(x_\alpha)_{\alpha\in\Lambda^{<b}}$ is not summable (or vice versa)?

The answer is yes in both directions, with completely explicit real matrix algebras. We use the source convention that ordered products accumulate new factors on the left.

2 Proof intuition

Noncommutativity lets a short block cancel additively while retaining a second-order multiplicative drift. For nilpotent matrix units this drift is exact: a four-factor commutator equals $I - t^2W$, with no higher-order remainder. Repeating it at $t = n^{-1/2}$ produces harmonic product drift but zero additive drift. Appending the inverse drift factor makes every block product the identity while leaving a harmonic additive sum. Embedding the resulting sequences as shrinking steps transfers these two elementary block calculations to the two requested integrability separations.

3 The sequence-to-step device

Let $(x_j)_{j\geq 1}$ be a sequence in a finite-dimensional unital Banach algebra E, and put

$$\alpha_j = 1 - 2^{1-j}, \qquad \Lambda = \{\alpha_j : j \geq 1\} \cup \{1\}.$$

Then $\Lambda \subset [0,1]$ is well ordered, its only limit element is 1, and $\alpha_{j+1} - \alpha_j = 2^{-j}$. Define the step mapping

$$F_x(t) = 2^j x_j \quad (t \in [\alpha_j, \alpha_{j+1})), \qquad F_x(1) = 0. \tag{1}$$

Thus the additive increment on the j-th step is exactly x_j, and its exponential increment is e^{x_j}.

Lemma 1. *Assume $x_j \to 0$.*

1. *If $P_N = \mathrm{e}^{x_N} \cdots \mathrm{e}^{x_1}$ converges to an invertible P, then F_x is strongly Kurzweil product integrable and its product integral is P.*
2. *The mapping F_x is Henstock–Kurzweil integrable exactly when $\sum_j x_j$ converges.*
3. *If (P_N) does not converge, then F_x is not Kurzweil product integrable.*

Proof. For (1), set $P_0 = I$ and, for $t \in [\alpha_j, \alpha_{j+1})$, define

$$W(t) = \exp\big((t - \alpha_j)2^j x_j\big)P_{j-1}, \qquad W(1) = P.$$

At every finite step boundary the two definitions agree. Moreover, $P_{j-1} \to P$ and $\|(t-\alpha_j)2^j x_j\| \le \|x_j\| \to 0$; hence $W(t) \to P$ as $t \uparrow 1$. Thus W is continuous and invertible, and $W'(t) = F_x(t)W(t)$ off the countable set of step boundaries. The continuous fundamental-matrix criterion in source Theorem 4.1 gives strong Kurzweil product integrability.

Statement (2) is the standard well-ordered-step criterion used by the source (its cited Proposition 3.1): the endpoint values of the additive primitive are $\sum_{j=1}^{N} x_j$. In finite dimension this is also immediate from continuity of the indefinite Henstock–Kurzweil integral and the converse construction of the piecewise-linear primitive. Finally, a product-integrable mapping with countably many discontinuities has a continuous indefinite product integral by source Theorem 4.1. Its values at α_{N+1} are P_N, proving (3). ☐

4 Exact nilpotent commutator blocks

Work first in $E = M_3(\mathbb{R})$, with the operator norm, and write

$$U = E_{12}, \qquad V = E_{23}, \qquad W = E_{13}.$$

Then $UV = W$, $VU = 0$, and every product involving W and one of U, V, W is zero. Since $U^2 = V^2 = W^2 = 0$, direct multiplication gives the exact identity

$$\mathrm{e}^{-tV}\mathrm{e}^{-tU}\mathrm{e}^{tV}\mathrm{e}^{tU} = I - t^2 W, \qquad \mathrm{e}^{t^2 W}(I - t^2 W) = I. \tag{2}$$

Theorem 2. *There are two explicit $M_3(\mathbb{R})$-valued step mappings on $[0,1]$, each with the well-ordered step set Λ above, such that:*

1. *the first is strongly Kurzweil product integrable, with product integral I, but is not Henstock–Kurzweil integrable;*
2. *the second is Henstock–Kurzweil integrable, with integral 0, but is not Kurzweil product integrable.*

Proof. Put $t_n = n^{-1/2}$. For the first mapping, concatenate the five-term blocks

$$t_nU, \quad t_nV, \quad -t_nU, \quad -t_nV, \quad t_n^2 W \qquad (n = 1, 2, \ldots) \tag{3}$$

into a sequence (x_j), and use (1). Each block has additive sum $t_n^2 W = W/n$. Consequently, at the block endpoints the additive partial sums are $H_N W$, where $H_N = \sum_{n=1}^{N} 1/n$, and hence the series does not converge.

In the ordered exponential product, however, (2) shows that every five-term block has product

$$\mathrm{e}^{t_n^2 W}\mathrm{e}^{-t_nV}\mathrm{e}^{-t_nU}\mathrm{e}^{t_nV}\mathrm{e}^{t_nU} = I.$$

The partial products inside the n-th block tend uniformly to I, because all five increments tend to zero. Thus the full sequence of partial products converges to I. Lemma 1 proves (1).

For the second mapping, concatenate only the first four terms in (3), obtaining a sequence (y_j). Every additive block sum is zero, and the largest norm of an intra-block additive partial sum is $O(t_n)$. Therefore $\sum_j y_j = 0$. On the other hand, the exponential product of block n is $I - W/n$. Since $W^2 = 0$, the product through the first N blocks is exactly

$$(I - W/N) \cdots (I - W) = I - H_N W,$$

which diverges in norm. Lemma 1 proves (2). ☐

5 Dimension-two refinement

The first separation can be realized in the 2×2 algebra highlighted by the source. Let $A = E_{12}$, $B = E_{21}$, and

$$C(t) = \mathrm{e}^{-tB}\mathrm{e}^{-tA}\mathrm{e}^{tB}\mathrm{e}^{tA} = \begin{pmatrix} 1-t^2 & -t^3 \\ t^3 & 1+t^2+t^4 \end{pmatrix}.$$

We have $\det C(t) = 1$ and $\operatorname{tr} C(t) = 2 + t^4 > 2$, so $C(t)$ has positive eigenvalues and a real principal logarithm. Set

$$s(t) = \operatorname{arcosh}(1 + t^4/2), \qquad D(t) = C(t) - (1 + t^4/2)I.$$

Since $D(t)^2 = \sinh^2(s(t))I$,

$$\log C(t) = \frac{s(t)}{\sinh s(t)} D(t). \tag{4}$$

Proposition 3. *There is an $M_2(\mathbb{R})$-valued step mapping with well-ordered steps which is strongly Kurzweil product integrable but not Henstock–Kurzweil integrable.*

Proof. For $t_n = n^{-1/2}$, concatenate the blocks

$$t_n A, \quad t_n B, \quad -t_n A, \quad -t_n B, \quad -\log C(t_n).$$

Their ordered exponential products are exactly $C(t_n)^{-1}C(t_n) = I$, and every increment tends to zero. Hence the partial products converge to I.

The first four additive terms cancel. Apply the linear functional $\ell(X) = (X_{11} - X_{22})/2$. Formula (4) gives

$$\ell(-\log C(t)) = \frac{s(t)}{\sinh s(t)} \left(t^2 + \frac{t^4}{2} \right).$$

The ratio $s/\sinh s$ tends to 1 as $s \to 0$, so for all sufficiently large n, the displayed quantity is at least $1/(2n)$. The additive series therefore diverges. Apply Lemma 1. $\square$

6 A source-equivalence caveat

The source calls its sequence formulation equivalent to the step-mapping question. The proof of the converse in its Theorem 4.2 uses $\exp(x_\alpha) \to I$ to infer $x_\alpha = \log(\exp x_\alpha) \to 0$. This requires eventual membership in a fixed logarithm chart and is false in general. For

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad x_j = 2\pi J,$$

all exponentials equal I, although $x_j \not\to 0$. The corresponding step mapping from (1) is not product integrable: its would-be fundamental matrix executes a full rotation on every interval and has no limit at 1.

This caveat does not affect Theorem 2 or Proposition 3. All their increments tend to zero, and product integrability was proved from an explicitly continuous fundamental matrix.

7 Verification, scope, and novelty

The exact identities in (2) and the algebra behind (4) were independently checked by the included SymPy script. The proof answers the first source question in full and, in $M_3(\mathbb{R})$, in both directions. It does not address the three other open bullets on PDF page 44.

A bounded search of the run indexes and of exact-question, title, product-integral, multipliability, matrix-product, correction, and erratum phrases found no later answer or matching construction through 17 August 2026. The source has few indexed citations, and a 2016 chapter by the same authors was the closest follow-up. Novelty is therefore plausible but should be checked by a specialist. Human review should focus on the source's precise strong/ordinary conventions and on the use of its Theorem 4.1 in Lemma 1.

References

[1] S. Heikkilä and A. Slavík, *On summability, multipliability, product integrability, and parallel translation*, arXiv:1410.1845; J. Math. Anal. Appl. 433 (2016), 887–934.

[2] A. Slavík, *Kurzweil and McShane product integration in Banach algebras*, J. Math. Anal. Appl. 424 (2015), 748–773.